Q1. If $N = 1$

$a = qb$ where $a \neq b$ and $\gcd = q$

$b \geq 1$ since $b \in \mathbb{N}$ and $a \geq 2$ since $a \neq b$ so

$a \geq F_{N+2}$ and $b \geq F_{N+1}$ is true for $N = 1$.

Suppose the algorithm takes n divisions to compute the gcd and $a \geq F_{n+2}$, $b \geq F_{n+1}$

Now suppose the algorithm takes $n+1$ divisions

From the first division:

$$a = q_1 b + r_1$$

Now, the algorithm takes n divisions to find the gcd of b and r_1

By the inductive hypothesis, $b \geq F_{n+2}$, and $r_1 \geq F_{n+1}$

Then $a \geq q_1 F_{n+2} + F_{n+1}$

The minimum q_1 can be is 1 so

$a \geq F_{n+2} + F_{n+1}$

$a \geq F_{n+3}$

so $a \geq F_{(n+1)+2}$ and $b \geq F_{(n+1)+1}$

Therefore if the hypothesis holds for $N = n$, it also holds for $N = n+1$.

Together with the case $N = 1$, we have shown that the statement holds for all $N \in \mathbb{N}$.